Hrushikesh's Bachelor's Thesis

Performance of batteries under extreme loading conditions: Data Driven approach.

Objectives:

• To simulate the electro-thermal response of lithium-ion batteries under internal short-circuit conditions for analysis of unsafe behaviour and generation of high-fidelity datasets

• To develop a data-driven framework for predicting thermal runaway risk using inconsistencies in measurable operating parameters as proxies for internal faults

Methodology:

- Implemented a semi-empirical NTGK electro-thermal model in ANSYS Fluent (MSMD) to simulate lithium-ion battery behaviour, modelling internal short circuits via localized volumetric resistance to capture hotspot formation.
- Performed a parametric study by varying C-rate, short circuit severity, ambient temperature, and heat transfer coefficient across charging and discharging conditions.
- Generated a large simulation dataset (~1000 cases) capturing temperature, voltage, and SOC evolution under diverse operating and fault scenarios.
- Engineered physics-informed features, particularly SOC slope deviation from nominal behaviour, to indirectly characterize internal short circuits using BMS measurable parameters.
    - Inputs: SOC deviation, Ambient conditions, C Rate
    - Output: Temperature (Temperature range classifies risk level)
- Developed supervised ML models for regression (Random Forest, XGBoost, SVR, ANN) to predict maximum battery temperature from operating conditions and engineered features. Benchmarked them against each other to identify best possible model to be used.
- Defined temperature-based risk thresholds and evaluated model performance using RMSE and classification matrices to assess thermal runaway risk prediction accuracy.

Conclusion

- Developed a simulation-based data driven framework for battery safety prediction, combining simulation data to derive relevant insights
- A robust simulation framework accurately predicts battery response, and captures relevant temporal data of SOC, temperature, C-rate and Voltage.
- ML model enables early detection of internal short circuits using measurable BMS signals (SOC trends, ambient conditions and battery status)
- Provides a foundation for integration into battery management systems for safety monitoring

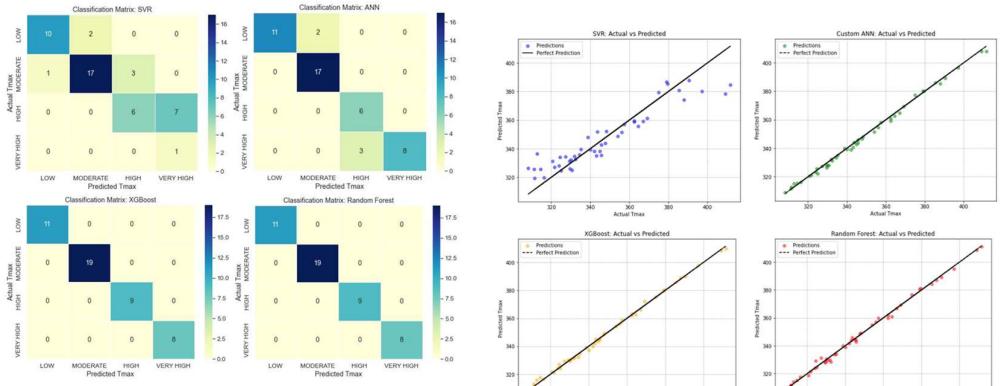

Fig 37: Classification Matrices for all the given models for the discharge case